

MUHAMMAD ABDULLAH

Data Analyst | Data Scientist | Python Developer

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in Python, SQL, R, data analysis, machine learning, and full-stack application development. Skilled in data cleaning, exploratory data analysis, statistical testing, predictive modeling, and clustering, with proven ability to build and deploy end-to-end machine learning applications. Developed Biz Ledger, a full-stack financial ERP system using React, TypeScript, FastAPI, SQLAlchemy, and PostgreSQL, including accounting workflows, financial reporting, authentication, and role-based access control. Certified IT Specialist in Data Analytics (Certiport/Pearson VUE). Seeking an entry-level position in Data Analytics, Data Science, Python Development, or related software/data roles.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, R

Data Analysis & Machine Learning: Pandas, NumPy, Scikit-learn, Feature Engineering, Classification, Regression, Random Forest, K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA), Model Evaluation, Cross-Validation

Statistics: Descriptive Statistics, Probability, Hypothesis Testing, Statistical Significance Testing

Data Visualization & Business Intelligence: Power BI, Matplotlib, ggplot2, Microsoft Excel (Pivot Tables, VLOOKUP)

Web Development & Backend: FastAPI, SQLAlchemy, REST APIs, PostgreSQL, React, TypeScript, JWT Authentication

Tools & Deployment: Streamlit, Git, GitHub, Jupyter Notebook, Google Colab

PROJECTS

Biz Ledger — Full-Stack Financial ERP System | *Personal Project, 2026*

React, TypeScript, FastAPI, Python, PostgreSQL, SQLAlchemy, JWT

- Built a full-stack financial ERP system covering customers, suppliers, invoices, bills, payments, banking, journal entries, and financial reporting.
- Implemented double-entry accounting workflows with debit/credit validation across invoices, bills, payments, bank transactions, transfers, and journal entries.
- Developed General Ledger, Trial Balance, Profit & Loss, Balance Sheet, Cash Flow, and bank reconciliation modules with automated accounting data validation.
- Implemented JWT authentication, role-based permissions, company-level data isolation, password reset, and audit logging.
- Designed professional A4 invoice, bill, and payment documents with browser-based print support.
- Built the frontend in React and TypeScript with a FastAPI/SQLAlchemy backend and PostgreSQL database.

Customer Churn Prediction — End-to-End Machine Learning Application | *Academic Project, 2026*

Python, Pandas, NumPy, Scikit-learn, Streamlit

- Cleaned and analyzed a 7,043-record telecom customer churn dataset, including missing-value handling, exploratory data analysis, feature preparation, and categorical encoding.
- Engineered features and trained classification models to predict customer churn; Random Forest model achieved approximately 0.64 F1-score on the evaluation set.
- Built an end-to-end Scikit-learn preprocessing and modeling pipeline using ColumnTransformer, StandardScaler, OneHotEncoder, and Random Forest.
- Deployed the trained model as an interactive Streamlit web application for customer-level churn prediction and model insights.
- Created data visualizations to communicate churn patterns and key predictive features to non-technical stakeholders.

Customer Segmentation Using K-Means Clustering | *Academic Project, 2024*

R, Cluster, factoextra, ggplot2

- Applied K-Means clustering to segment customers based on purchasing behavior and demographic characteristics.
- Determined the optimal number of clusters using the Elbow Method and Silhouette Analysis.

- Visualized cluster structures and analyzed segment characteristics to support targeted marketing strategies.

Hierarchical Clustering & Anomaly Analysis | *Academic Project, 2023*

R, PCA, Hierarchical Clustering, Statistical Analysis

- Applied agglomerative hierarchical clustering to multivariable datasets to identify meaningful groups and structural patterns.
- Used Principal Component Analysis (PCA) for dimensionality reduction and data visualization.
- Analyzed distance-based patterns and dendrogram structures to identify anomalies and cluster characteristics.

TRAINING & JOB SIMULATIONS

Data Science Job Simulation — British Airways (Forage) | *2024*

- Performed data preparation and exploratory data analysis using Python and Pandas on a simulated airline customer dataset.
- Built a classification workflow using Scikit-learn to analyze customer booking behavior.
- Applied statistical analysis to identify relevant patterns in customer and transactional data.
- Translated analytical findings into business recommendations through structured reporting and visualization.

Data Analytics Job Simulation — Quantum (Forage) | *2024*

- Performed data preparation, customer and transaction analysis, and exploratory data analysis using Python.
- Analyzed purchasing behavior and customer segments to identify commercially relevant patterns.
- Communicated findings through structured analysis and business-oriented recommendations.

EDUCATION

BS Computer Science | *Government College University Faisalabad, 2020 – 2026*

CGPA: 3.19 / 4.00

CERTIFICATIONS

- IT Specialist – Data Analytics, Certiport / Pearson VUE, 2024
- Data Analytics Job Simulation, Quantum — Forage, 2024
- Data Science Job Simulation, British Airways — Forage, 2024